

The Ambient AI Lab

AI for Scientific Practice

Will Handley

[<wh260@cam.ac.uk>](mailto:wh260@cam.ac.uk)

Royal Society University Research Fellow
Institute of Astronomy, University of Cambridge
Kavli Institute for Cosmology, Cambridge
Gonville & Caius College
willhandley.co.uk/talks

7th September 2026



UNIVERSITY OF
CAMBRIDGE



Scientists versus agents

Nature Astronomy Comments, 2026: Institute of Astronomy, Cambridge



Large language models are not the problem

Hiranya V. Peiris

Nature Astronomy, 3 April 2026

[[doi:10.1038/s41550-026-02837-2](https://doi.org/10.1038/s41550-026-02837-2)] [2604.22071]



Agentic research is oxymoronic

Natalie B. Hogg

Nature Astronomy, 31 August 2026

[[doi:10.1038/s41550-026-02954-y](https://doi.org/10.1038/s41550-026-02954-y)] [2608.31161]

ANTHROPIC

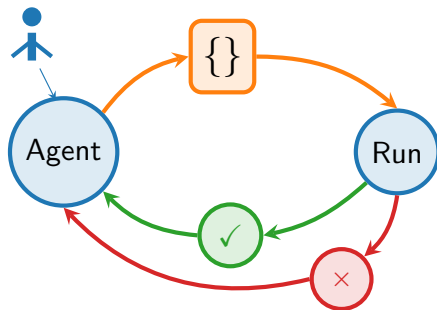
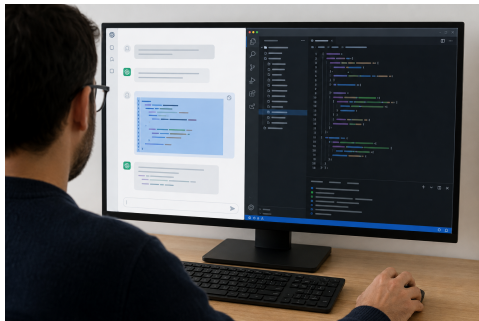


An agent is ChatGPT that can edit files and run commands

This closes the scientific loop.

If you get too skilled at chat-based AI, you become a **copy-paste-run-debug monkey**.

Agentic systems are what happens if you let ChatGPT **edit files, run commands, and inspect the behaviour**.



Case Study 1: Agents do coding grunt work at scale

emcee → BlackJAX

Will Handley, October 2025
Manual: 2 months. LLM: 15 min.
Matches reference emcee numerically.
Written for a Monte Carlo figure in a talk.

jaxwavelets

Will Handley
PyWavelets reimplemented in JAX.
 10^{-14} agreement, 1,177 tests.
4.7 hours' active computation.
Available on PyPI.

Astro Legacy Archive

Agents audited 2,130 Hugging Face datasets.
Archive proposal under review.

jax-bandflux



Sam Leeney
[[2504.08081](#)]

SALT3 bandflux on GPU: $\sim 100\times$.
Differentiable; initial draft in ~ 3 h.

JAX-SNANA



Matt Grayling
SN light-curve pipeline
rewritten in JAX

~ 200 simulations/s on CPU
 $> 10,000$ /s on A100.
DES Y5 agreement: < 1 millimag.

Bug Hunting

Sinah Legner: CAMB torsion
 N_{eff} counted twice; an evolution
pathway for massive neutrinos was
missing.

Discrete k in CLASS:
FCB expansion had a sign error.

Covariance in minutes

Natalie Hogg: 24 h $\rightarrow$ 1–2 min.

HPC jobs and meeting scripts

Submit and monitor HPC jobs.
Write and run scripts in meetings.

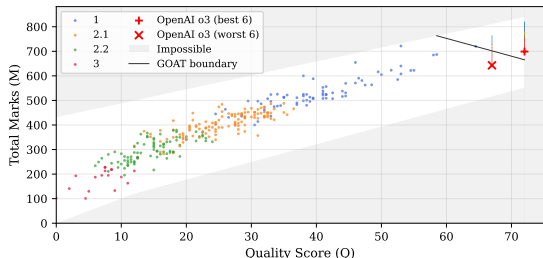
Case Study 2: Artificial intelligence in examinations

May 2025

I got OpenAI's o3 to take all four Part II Astrophysics papers.

Toby Lovick transcribed the answers by hand into examination booklets.

I slipped them amongst real scripts for blind marking.



April 2026

I could not use Claude Code or Codex: uploading confidential papers risked exposing the material to students.

Qwen and Gemma ran through Alan on my own machine, disconnected from the internet, with two RTX 6000 GPUs.

They helped assemble the papers and identified typographical and substantive errors overlooked by the examiners.

Case Study 3: Live transcripts at SBI4GALEV

Simulation Based Inference for Galaxy Evolution, 23–26 June 2026



Toby Lovick
Built the system

Current slide + live transcript

↓ talk ends

Summary + three suggested questions



Chris Lovell
Operated and developed it
during the meeting

One participant declined recording but wanted the transcript. We could not separate the two.

Publication required an explicit opt-in.
53 talks captured, 39 in the public archive.

Open in a browser on the local network. No login.
Anonymous votes on the questions.

People particularly valued being able to read along.

All local: NeMo and Gemma 4 31B on turing, our GPU machine. No cloud processing.

Public archive

[[github:sbi-galev/ambient-ai](https://github.com/sbi-galev/ambient-ai)]

An agent cannot do all the science in a paper

My rule of thumb: about 80% of the work can be handed over

80%

Grunt work for agents

20%

Human
judgement

We decide what to calculate, and what to make of the answer: judgement and taste.

Those choices were human in every example you've just seen.

A paper produced by agents from start to finish leaves that part undone.

In principle: 5× faster, 5× more fun.

AI in our research lab today

Claude Code, Codex, Otter and our GitHub repositories

Otter and `mdrecord`

Otter $\xrightarrow{\text{nightly}}$ `mdrecord`

- ▶ About 1,000 meetings saved in 18 months
- ▶ This talk began with a search through our meeting transcripts

Our 2025 tools: [[github:handley-lab/mcp-handley-lab](https://github.com/handley-lab/mcp-handley-lab)]

Claude Code and Codex

Code, slides and tasks during meetings

- ▶ Nikku Madhusudhan: K2-18b retrieval code assembled as we talked
- ▶ James Alvey and Thomas Spieksma: requested Einstein Telescope figure and source quotations displayed within 15 s

The Ambient AI Lab we're building

Cosmos grant: \$5,000 for 90 days of development with Toby Lovick

- ▶ Frontier tools expect us at a computer, and our work follows.
- ▶ We work best together at a board, in meetings and at conferences
- ▶ Bring agents into the room before we become “digitally rendered meat-backed slack bots”
- ▶ Record speech and the board without changing how people work
- ▶ Opt-in recording. Our data stays on our local machines.
- ▶ More supervisors in Lent

Figure request for Will

Proposed room layout for the Cosmos project: people at the board, microphones, camera, display and local storage. Show how equipment fits the room; distinguish existing hardware from planned additions.

Cosmos Institute:

“Instrumenting the Unrecorded Half of Science”

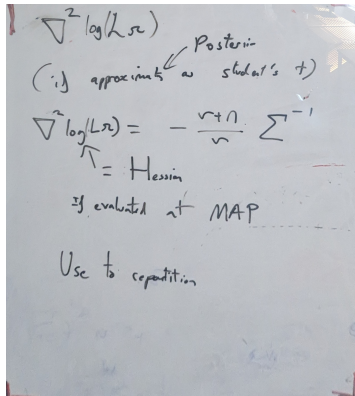
In development: speech → Whisper → Claude Code

Checking the whiteboard

Toby Lovick



Photograph → Gemini 3.1 → LaTeX; then the live loop



$$\nabla^2 \log(L\Omega)$$

Posterior

(if approximate as student's t)

$$\nabla^2 \log(L\Omega) = -\frac{v+n}{v} \Sigma^{-1}$$

↑
= Hessian

If evaluated at MAP

Use to repartition

- ▶ Live: Pi camera → turing → projector
- ▶ Ask “where is this wrong?”

We can run the lab's agents on our own GPUs

I would not assume the token subsidy is permanent

\$1,000/day

My usage, comparable to my ERC funding.

The service bundles a harness with subsidised tokens.

If the terms become unacceptable, or we have to pay the full cost, we can move the work.

For this scripting and analysis, we can write the harness and run the models ourselves.

Already local: April's exam preparation with Qwen and Gemma, and Toby's Gemma whiteboard demonstration.

Conclusions

handley-lab.co.uk

- ▶ **I agree with Hiranya V. Peiris and Natalie B. Hogg.**

I use agents for much of my working day.

- ▶ **Agents can take on most of the labour.**

My rule of thumb is 80%. The other 20% needs our judgement and taste.
Generating a complete paper does not supply them.

Ambient, like electricity

There when we need it, in the room where we work.

It does not choose the problem or judge the result.